



E0098

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Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
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Machine Learning

Exam: IN2064 / Retake

Date: Tuesday 31st March, 2026

Examiner: Prof. Dr. Stephan Günnemann

Time: 14:00 – 16:00

P 1

P 2

P 3

P 4

P 5

P 6

P 7

P 8

P 9

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Working instructions

- This exam consists of **16 pages** with a total of **9 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 40 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - Two-sided DIN A4 sheet of handwritten notes (a print of digitally handwritten notes is allowed)
- **No other material (e.g. books, cell phones, calculators) is allowed!**
- Physically turn off all electronic devices, put them into your bag and close the bag.
- There is scratch paper at the end of the exam.
- Write your answers only in the provided solution boxes or the scratch paper.
- If you solve a task on the scratch paper, clearly reference it in the main solution box.
- All sheets (including scratch paper) have to be returned at the end.
- **Only use a black or a blue pen (no pencils, red or greens pens!)**
- **For problems that say “Justify your answer” you only get points if you provide a valid explanation.**
- **For problems that say “Derive” you only get points if you provide a valid mathematical derivation.**
- **For problems that say “Prove” you only get points if you provide a valid mathematical proof.**
- If a problem does not say “Justify your answer”, “Derive” or “Prove”, it is sufficient to only provide the correct answer.

Left room from _____ to _____ / Early submission at _____



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Problem 1 Weighted k-Nearest Neighbors (k-NN) (3 credits)

Recall that the ℓ_p -distance between two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ is defined as

$$d_p(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p = \left(\sum_{i=1}^d |x_i - y_i|^p \right)^{1/p},$$

where the ℓ_∞ -distance is based on the maximum norm and defined as $d_\infty(\mathbf{x}, \mathbf{y}) = \max_{i=1, \dots, d} |x_i - y_i|$.

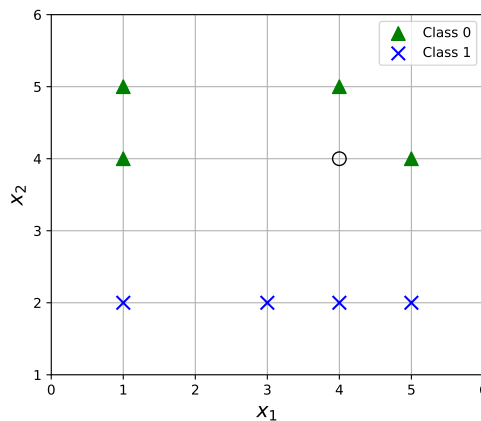
Moreover, the weighted k-NN classifier is defined as

$$\hat{y} = \arg \max_{c \in \mathcal{C}} \frac{1}{Z} \sum_{\mathbf{x}_i \in \mathcal{N}_k(\mathbf{x})} w_i \cdot \mathbb{I}(y_i = c) \quad \text{with} \quad \frac{1}{Z} = \sum_{\mathbf{x}_i \in \mathcal{N}_k(\mathbf{x})} w_i.$$

A potential issue in k-NN classification arises when multiple training points are equally distant to the point to be classified, making them all eligible as the k-th nearest neighbor. This can lead to ambiguity in the classification decision, as it is unclear which points should be considered in the majority vote.

One way to mitigate this is to locally enlarge the neighborhood to include *all equally distant* k-th nearest neighbors. For example in 1-NN: If there is a tie between two points (both at the same distance), then you would enlarge the neighborhood to include both points, effectively using two neighbors for the 1-NN decision.

We want to apply this k-NN variant with locally enlarged neighborhood to this binary classification problem:



- 0 ☐ a) Classify the test point (marked by $\circ$) at coordinates (4, 4) using 3-NN with weights $w_i = 1$ for the d_1 and d_∞ distances. Justify your answer by showing the weighted votes for each class.

1 ☐





0
1

b) Classify the test point (marked by $\circ$) at coordinates (4, 4) using 3-NN with weights $w_i = \frac{1}{d_p(\mathbf{x}, \mathbf{x}_i)}$ for the d_1 and d_∞ distances. Justify your answer by showing the weighted votes for each class.

c) Consider any k-NN classifier with weights $w_i = \frac{1}{d_p(\mathbf{x}, \mathbf{x}_i)}$ for $p = 1$ and $k \leq |\mathcal{D}|$ for the dataset $\mathcal{D}$. You are an adversary who may insert one additional training point $\hat{\mathbf{x}}$ with the new label 2, with the goal of causing the test point ($\circ$) to be misclassified as this new third class. Describe where you would place the new training point to achieve this. Justify your answer.

0
1





Problem 2 Probabilistic Inference (4 credits)

Suppose we observe i.i.d. data $\mathcal{D} = \{x_1, \dots, x_N\}$, with $x_i \in \mathbb{R}$, sampled from a shifted exponential distribution with unknown location parameter $\theta \in \mathbb{R}$:

$$p(x|\theta) = \begin{cases} \lambda \exp(-\lambda(x - \theta)) & \text{if } x \geq \theta \\ 0 & \text{otherwise,} \end{cases}$$

where $\lambda > 0$ is known.

0 ☐

1 ☐

2 ☐

a) For which values of θ is the data likelihood $p(\mathcal{D}|\theta) = \prod_{i=1}^N p(x_i|\theta)$ zero? And for which values is it non-zero? Show your work.





b) Justify that the data likelihood is monotonically increasing in θ in the region, where it is non-zero.

Large empty box for answer to question b)

c) What is then the maximum likelihood estimate $\theta_{\text{MLE}} = \arg \max_{\theta} p(\mathcal{D}|\theta)$? Show your work.

Large empty box for answer to question c)





Problem 3 Linear Regression (3 credits)

We want to perform ridge regression on a dataset consisting of N samples $\mathbf{x}_i \in \mathbb{R}^D$ ($D > 1$) with corresponding targets $y_i \in \mathbb{R}$, represented compactly as $\mathbf{X} \in \mathbb{R}^{N \times D}$ and $\mathbf{y} \in \mathbb{R}^N$. The closed-form solution for ridge regression is given by $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$, where $\lambda > 0$ is the regularization strength.

- 0 ☐
- 1 ☐
- 2 ☐

a) Show that $\mathbf{w}^*$ can be equivalently written as $\mathbf{w}^* = \mathbf{X}^T (\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})^{-1} \mathbf{y}$, which is called the dual form of ridge regression.

Hint: Use the fact that $(\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})(\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})^{-1} = \mathbf{I}$.

- 0 ☐
- 1 ☐

b) When is the dual form useful from a computational perspective?





Problem 4 Linear Classification (4 credits)

Consider a binary generative classifier defined by class probabilities

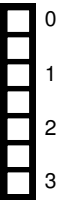
$$p(y = 0) = \pi_0, \quad p(y = 1) = 1 - \pi_0,$$

and class-conditional densities given by Laplacian distributions

$$p(x|y = c) = \frac{1}{2b} \exp\left(-\frac{|x - \mu_c|}{b}\right),$$

where $x \in \mathbb{R}$ is the input feature, $\pi_0 \in [0, 1]$, $b > 0$, $c \in \{0, 1\}$, and $\mu_c \in \mathbb{R}$ are the class-conditional means.

a) Is the classifier linear in x ? Justify your answer.



b) Name one advantage of using a Laplace distribution instead of a Gaussian distribution for $p(x|y)$.





Problem 5 Convexity and the Hessian (6 credits)

Multiple choice scoring rule: Each sub-task is scored all-or-nothing. Full points are awarded if and only if exactly the correct set of options is selected. Any incorrect or missing selection yields 0 points for that sub-task.

a) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable. Which of the following are correct?

- ☐ f is convex if and only if $f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y}$ and $\lambda \in [0, 1]$.
- ☐ If f is convex, then every local minimum is a global minimum.
- ☐ If $\nabla^2 f(\mathbf{x})$ has all eigenvalues > 0 at a single point $\mathbf{x}_0$, then f is strictly convex.
- ☐ The sum of two convex functions is always convex.
- ☐ If f is convex and g is convex, then $f \circ g$ is convex.

b) Consider the function $f(x, y) = x^2 - 2xy + y^2 + x$. Which of the following are correct? (2 Points)

- ☐ f is strictly convex.
- ☐ f is convex but not strictly convex.
- ☐ The Hessian is positive semidefinite because all diagonal entries are positive.
- ☐ The Hessian is positive semidefinite because all eigenvalues are ≥ 0 .
- ☐ f has a unique global minimum.

c) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable and convex. A point $\mathbf{x}^*$ satisfies $\nabla f(\mathbf{x}^*) = \mathbf{0}$. Which of the following are correct?

- ☐ $\mathbf{x}^*$ is a global minimum of f .
- ☐ $\mathbf{x}^*$ is the unique global minimum of f .
- ☐ $\nabla^2 f(\mathbf{x}^*)$ must be positive definite.
- ☐ There may exist other points $\mathbf{x}' \neq \mathbf{x}^*$ with $f(\mathbf{x}') = f(\mathbf{x}^*)$.
- ☐ $\mathbf{x}^*$ is a saddle point.

d) Consider the function $h(x, y) = e^{x+y}$. Which of the following are correct? (2 Points)

- ☐ The Hessian of h has rank 2 everywhere.
- ☐ The Hessian of h has rank 1 everywhere.
- ☐ h is strictly convex.
- ☐ h is convex but not strictly convex.
- ☐ h has a global minimum.





Problem 6 Deep Learning (6 credits)

You are implementing a deep learning model to be trained with backpropagation. Let

$$z = Wx, \quad y = z + x, \quad s = y \odot y, \quad L = \sum_{i=1}^N s_i$$

where $x \in \mathbb{R}^N$, $W \in \mathbb{R}^{N \times N}$, $z, y, s \in \mathbb{R}^N$ and $L \in \mathbb{R}$. $\odot$ denotes elementwise multiplication. You have access to the following operations:

- `matmul(W, x)`: matrix-vector multiplication
- `add(u, v)`: elementwise addition
- `square(y)`: elementwise squaring
- `sum(s)`: summation over all entries

a) Draw the computational graph of the forward pass that computes L from W and x , using only the operations above.



b) Write the chain rule expressions for $\frac{\partial L}{\partial W}$ and $\frac{\partial L}{\partial x}$ based on your computational graph using all intermediate variables.



c) Which intermediate values in your computational graph must be cached during the forward pass in order to compute $\frac{\partial L}{\partial W}$ and $\frac{\partial L}{\partial x}$ during backpropagation without recomputation? **Justify your answer.**
Note: W and x are assumed to be available and do not need to be cached.





Problem 7 Support Vector Machines (3 credits)

Multiple choice scoring rule: Each sub-task is scored all-or-nothing. Full points are awarded if and only if exactly the correct set of options is selected. Any incorrect or missing selection yields 0 points for that sub-task.

Consider the soft-margin SVM primal problem:

$$\min_{\mathbf{w}, b, \xi} \quad \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad \forall i,$$

where $C > 0$ is a hyperparameter that controls the penalty for margin violations: each training point $\mathbf{x}_i$ that does not satisfy the margin constraint $y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1$ incurs a slack $\xi_i > 0$, and the total slack $\sum_i \xi_i$ is penalized in the objective with weight C . A larger C penalizes violations more heavily.

The corresponding dual problem has Lagrange multipliers α_i with $0 \leq \alpha_i \leq C$. At the optimum, the KKT conditions imply:

- $\alpha_i^* = 0$: $\mathbf{x}_i$ is correctly classified and lies outside or on the margin.
- $0 < \alpha_i^* < C$: $\mathbf{x}_i$ lies exactly on the margin boundary ($\xi_i^* = 0$).
- $\alpha_i^* = C$: $\mathbf{x}_i$ may violate the margin ($\xi_i^* \geq 0$).

For the special case $C \rightarrow \infty$, no margin violations are permitted and the problem reduces to the hard-margin SVM (which requires the data to be linearly separable).

a) A hard-margin SVM is trained on a linearly separable dataset $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ in $\mathbb{R}^2$, yielding optimal dual variables $\alpha_1^*, \dots, \alpha_N^*$. Which of the following modifications to the training set will *not* change the optimal decision boundary?

- ☐ Removing a training point with $\alpha_i^* = 0$.
- ☐ Adding a new positive-class point that is correctly classified and lies outside the margin.
- ☐ Moving a support vector closer to the decision boundary along its perpendicular.
- ☐ Adding a new negative-class point that lies exactly on the positive margin boundary, i.e. at $\mathbf{w}^{*\top} \mathbf{x} + b^* = +1$.
- ☐ Adding a new positive-class point that lies exactly on the decision boundary, i.e. at $\mathbf{w}^{*\top} \mathbf{x} + b^* = 0$.

b) A soft-margin SVM with $C = 10$ is trained. A particular training point $\mathbf{x}_j$ with label $y_j = +1$ has optimal dual variable $\alpha_j^* = 10$ and slack $\xi_j^* = 0.4$. Which statements about $\mathbf{x}_j$ are correct?

- ☐ $\mathbf{x}_j$ lies exactly on the margin boundary.
- ☐ $\mathbf{x}_j$ is correctly classified but lies strictly inside the margin.
- ☐ $\mathbf{x}_j$ is misclassified (on the wrong side of the decision boundary).
- ☐ $\mathbf{x}_j$ is not a support vector.
- ☐ $\mathbf{x}_j$ lies exactly on the decision boundary.

c) Two soft-margin SVMs are trained on the same non-separable dataset: one with $C = 0.1$ and one with $C = 1000$. Which statements are necessarily true?

- ☐ The $C = 0.1$ model has a wider margin (larger $\frac{2}{\|\mathbf{w}\|}$).
- ☐ The $C = 1000$ model has fewer support vectors.
- ☐ The $C = 0.1$ model has smaller total slack $\sum_i \xi_i^*$.
- ☐ Increasing C beyond 1000 will eventually achieve 100% training accuracy on this dataset.
- ☐ The $C = 1000$ model will always have higher test accuracy than the $C = 0.1$ model.



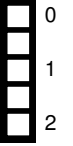


Problem 8 Singular Value Decomposition (5 credits)

A matrix $\mathbf{M} \in \mathbb{R}^{4 \times 3}$ records the ratings of 4 users for 3 movies. Its SVD is given by $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ with

$$\mathbf{U} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix}, \quad \mathbf{\Sigma} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad \mathbf{V}^T = \frac{1}{3} \begin{pmatrix} 2 & 1 & -2 \\ 2 & -2 & 1 \\ 1 & 2 & 2 \end{pmatrix}.$$

a) Derive the best rank-2 matrix approximation of $\mathbf{M}$ in terms of the Frobenius norm. You do not need to compute all entries and can instead derive its construction from $\mathbf{U}$, $\mathbf{\Sigma}$, and $\mathbf{V}$.





0 ☐

1 ☐

b) A new user rates the three movies as $\mathbf{x} = (0, 3, 0)^T$, where a rating of 0 indicates an unseen movie. Compute the representation of this user in the 2-dimensional concept space defined by the rank-2 approximation.

0 ☐

1 ☐

2 ☐

c) Reconstruct the predicted ratings for this user from the concept-space representation. Among the movies the user has not yet seen, which one does the model recommend?

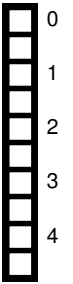




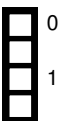
Problem 9 GMMs & EM (6 credits)

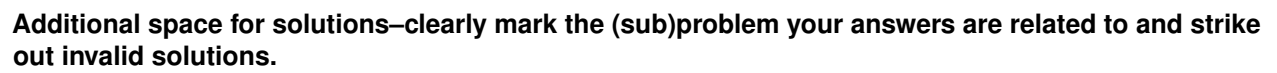
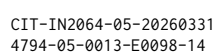
a) Consider a GMM with K components where all components share the same covariance matrix $\Sigma = \sigma^2 I \in \mathcal{R}^{D \times D}$. Let γ_{ik} denote the responsibility of component k for data point $\mathbf{x}_i$, computed in the E-step. Prove that in the limit $\sigma \rightarrow 0$, the responsibilities γ_{ik} converge to hard assignments, where each data point is assigned entirely to the component with the nearest mean.

Hint: Consider the ratio $\frac{\gamma_{ik^*}}{\gamma_{ik}}$ where $k^* = \arg \min_j \|\mathbf{x}_i - \mu_j\|^2$ and $k \neq k^*$.



b) How does this change the M-step? Explain how this limit case relates to the K-means algorithm. Justify your answer.



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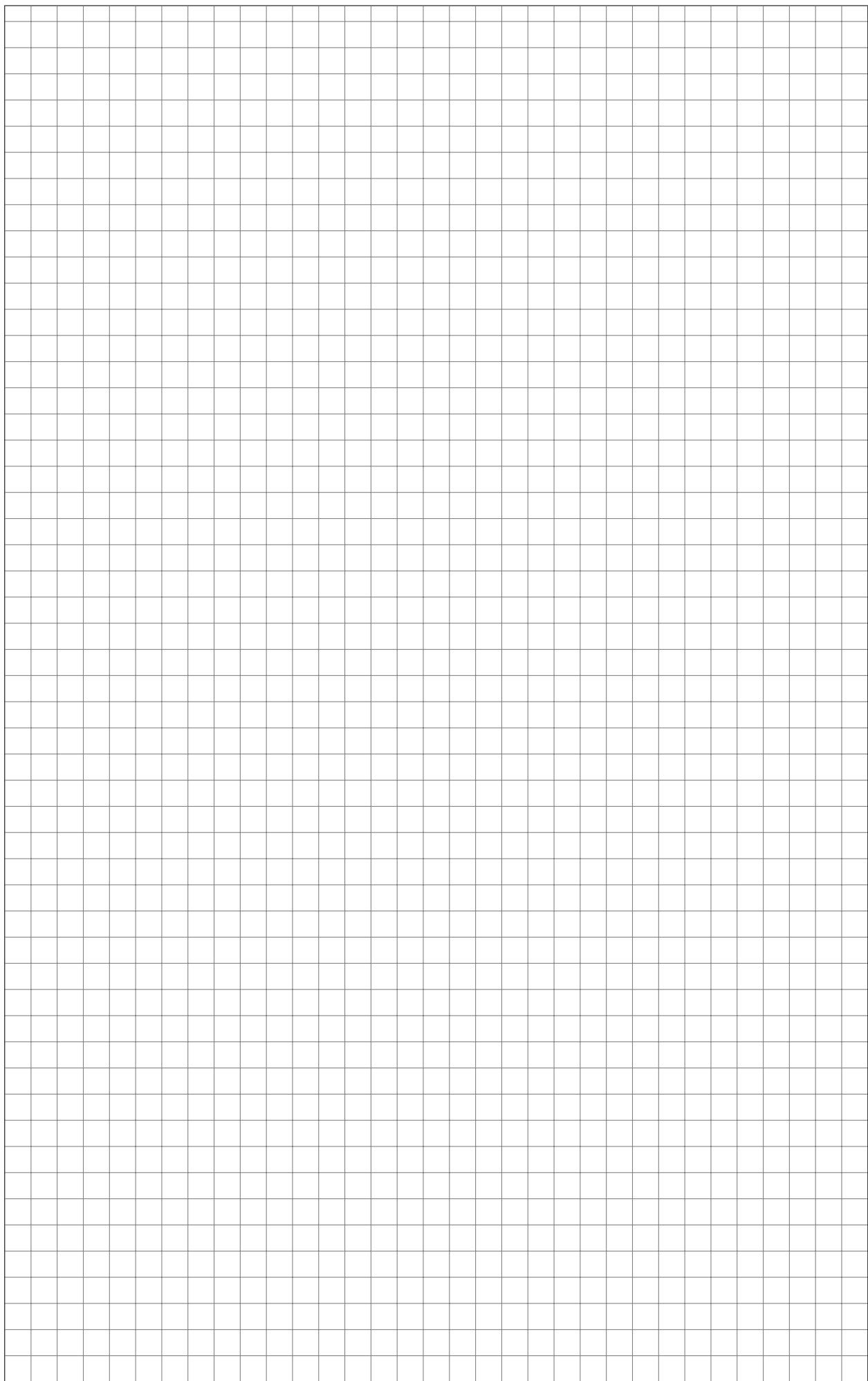


4794-05-0013-E0098-15



– Page 15 / 16 –

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